

Weekly Project Update: CHF Prediction for Lithium Battery Cooling

Critical Heat Flux (CHF) prediction using machine learning, PINNs, and external paper data

Short summary: This week we studied CHF, prepared a clean lookup-table dataset, compared many machine-learning models, tested honest high-pressure extrapolation, developed physics-informed neural networks, and started building a richer external test dataset. The main lesson is that a model can look excellent inside the known data range and still fail when it must predict new conditions.

1. Problem statement given for the project

- Critical Heat Flux (CHF) is the maximum heat flux that a cooling surface can remove before the boiling process changes suddenly and the surface can overheat.
- For lithium-battery cooling, the aim is to predict this safe heat-removal limit for different pressure, flow, quality, fluid, and geometry conditions.
- A reliable CHF prediction can reduce the number of expensive experiments and help us identify unsafe operating conditions before a thermal failure occurs.
- The first model is built from the 2006 Groeneveld water CHF table. This gives us a controlled baseline; the battery-specific and geometry-specific testing is being added through external papers and the CAD experiment.

2. Three research papers read first

Paper	Simple takeaway for this project
Applications of Machine Learning Methods for Boiling Modeling and Prediction	Boiling is affected by many coupled variables. ML can give high accuracy, but the result depends on the input features, model choice, and the range covered by the training data.

<i>The 2006 CHF Look-Up Table</i>	<i>The table gives CHF for a reference vertical 8 mm water-cooled tube as a function of pressure (P), mass flux (G), and thermodynamic quality (X).</i>
<i>CHF Theory and Post-CHF Heat Transfer</i>	<i>At CHF, the heated wall can lose liquid contact and a vapour layer can form. Heat transfer then reduces and wall temperature can rise sharply; pressure, flow, quality, geometry, and orientation affect the result.</i>

3. How the lookup-table data was extracted

- *The raw digitized Excel table was read using the project data-preparation script. Merged pressure cells were forward-filled, spacer rows and repeated headers were handled, and the entries were converted into one long table with columns P, G, X, and CHF.*
- *The raw grid contained 11,592 rows from 24 pressure levels, 21 mass-flux levels, and 23 quality levels. Exactly 504 rows had CHF = 0, and all were at X = 1.0. These are boundary placeholders for the all-steam condition, so they were removed.*
- *The final usable dataset has **11,088** rows. P is in kPa, G is in kg/m²/s, X is thermodynamic quality, and CHF is in kW/m².*
- *CHF ranges from about 15 to 44,338 kW/m². Because this range is very wide, smooth models were also trained on ln(CHF), and predictions were converted back before calculating the final scores.*

4. Split A, B, and C logic

<i>Split</i>	<i>How it was made</i>	<i>What it tests</i>
<i>A: Random 80/20</i>	<i>Randomly use 80% for training and 20% for testing; repeat with five seeds (0 to 4).</i>	<i>An optimistic interpolation test: nearby grid points appear in both sets, so it is the easiest split.</i>

<i>B: Interior pressure holdout</i>	<i>Hold out every fourth interior pressure level, while pressures on both sides remain in training.</i>	<i>A moderate interpolation test: the model fills missing pressure slices inside the known range.</i>
<i>C: High-pressure holdout</i>	<i>Train only on $P \leq 16,000$ kPa and test on $P = 17,000$ to $21,000$ kPa.</i>	<i>The honest extrapolation test: the model must predict beyond the highest pressure it saw during training.</i>

5. Baseline model results: R2 score

The table below shows the main comparison. Split A values are five-seed means where applicable. For stochastic Split C models, the verified multi-seed mean is shown. Values are rounded for readability.

<i>Model</i>	<i>Split A R2</i>	<i>Split B R2</i>	<i>Split C R2</i>
<i>Grid interpolation (raw)</i>	<i>about 1.000</i>	0.9990	0.8415
<i>Extra Trees (raw)</i>	0.9992	0.9989	0.4335
<i>Random Forest (raw/log)</i>	0.9987	0.9877	0.4147
<i>XGBoost (log)</i>	0.9987	0.9876	0.4307
<i>LightGBM (log)</i>	0.9989	0.9879	0.4295

<i>GPR, Matern 5/2 (log)</i>	0.9916	0.9955	0.6751
<i>KNN, k = 3 (log)</i>	0.9931	0.9906	0.4528
<i>Standard MLP (log, verified mean)</i>	0.9906	0.9886	0.6277 +/- 0.0716
<i>Polynomial Ridge, degree 2 (log)</i>	0.8467	0.8707	0.7547
<i>Polynomial Ridge, degree 4 (log)</i>	0.9508	0.9471	0.6181
<i>Linear regression (log)</i>	0.8343	0.8469	-1.0786

Main result: tree models score close to 1.0 on interpolation, but their scores fall to about 0.43 on Split C because tree leaves cannot continue a pressure trend outside the training range. Grid interpolation is the strongest physical baseline, while degree-2 log-polynomial regression is the strongest simple trained ML baseline.

6. Four approaches tried to improve Split C

Approach	Split C result	What went wrong
<i>Physics-basis features with Ridge</i>	<i>R2 = -14.95 (raw degree 2); log versions were even worse.</i>	<i>New physics terms grew too strongly outside the training range and caused feature explosion.</i>

<i>Residual learning on a hybrid Biasi + Zuber correlation</i>	<i>R2 = -1.31 for the MLP residual model.</i>	<i>The residual error changed shape beyond 16,000 kPa, so the correction did not extrapolate.</i>
<i>Physics-penalty MLP with collocation points</i>	<i>R2 = 0.8484 for one seed.</i>	<i>This was a single-seed result; later verification showed strong seed sensitivity, so it was not used as the headline result.</i>
<i>Pressure-gated tree + smooth blend</i>	<i>R2 = 0.8547 for one seed; verified log-MLP blend mean = 0.6284.</i>	<i>The high score was a seed artifact; after multi-seed checking it was retracted.</i>

7. Ensemble model and the better Split C results

Model or ensemble	Split C R2	Comment
<i>Grid interpolation, raw</i>	0.8415	<i>Best direct table-based physical baseline.</i>
<i>Polynomial Ridge, degree 2, log</i>	0.7547	<i>Best deterministic trained ML model.</i>
<i>GPR, log</i>	0.6751	<i>Smooth model, but less reliable than the polynomial on the edge test.</i>
<i>Standard MLP, log, verified ensemble/mean</i>	0.6277	<i>Better than tree models on average, but still variable across seeds.</i>

<i>Residual-on-GridInterp PINN-style ensemble, SiLU</i>	0.8474	<i>Current best result; MAPE = 20.44%. It slightly improves the grid baseline.</i>
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The important idea was to let the lookup table provide the main physical trend and train the neural network only to learn a correction. The final prediction was:

$$\text{CHF prediction} = \text{GridInterp base} \times \exp(\text{neural correction})$$

This residual ensemble was trained with several seeds and the predictions were averaged in log space, which reduced random variation.

8. PINN formulation

A PINN is a neural network that learns from data while also receiving soft checks based on known physical behaviour. The output was based on $\log(\text{CHF})$, with P , G , and X standardized using training data only.

Initial architecture: $[P, G, X] \rightarrow 128 \text{ Tanh} \rightarrow 128 \text{ Tanh} \rightarrow 64 \text{ Tanh} \rightarrow \log(\text{CHF})$. The later residual model used a smaller $[16, 8]$ network with SiLU because the lookup-table base already carried most of the trend.

Loss function used:

$$L_{\text{total}} = L_{\text{data}} + \lambda_{\text{mono}} L_{\text{mono}} + \lambda_{\text{zuber}} L_{\text{zuber}} + \lambda_{\text{pos}} L_{\text{pos}}$$

- **L_{data} :** mean-squared error between predicted and true normalized $\log(\text{CHF})$.
- **L_{mono} :** penalty when the predicted CHF increases with quality X where the expected trend is decreasing.
- **L_{zuber} :** penalty when the pressure slope disagrees with the sign of the Zuber pool-boiling pressure trend.
- **L_{pos} :** penalty for negative predictions; later residual experiments used a positive exponential output instead.

9. Grid search, hyperparameter tuning, and Modal GPU training

The first PINN search was run on Modal using NVIDIA A10G GPUs. The real Split C test set was kept separate; the later PLAN used a proxy training/validation pressure band for choosing settings.

Stage	Main parameter change	Split C result	Interpretation
<i>Initial PINN grid search</i>	<i>Architectures [64,32], [128,64,32], [128,128,64]; lambda values; learning rates 0.001/0.0005; 256/512 collocation points; 5 seeds.</i>	Mean R2 = 0.8123; <i>MAPE = 32.87%</i>	<i>The best initial PINN configuration was 128 x 128 x 64 with lambda_mono = 0.1, lambda_zuber = 0.3, learning rate 0.0005, and 512 collocation points.</i>
<i>Tier 1 search</i>	<i>Proxy split, mini-batches, no separate positivity penalty, direct log model, [16,8] Tanh, multi-seed ensemble.</i>	<i>R2 = 0.7316;</i> <i>MAPE = 37.76%</i>	<i>More honest training procedure, but the small direct network was not strong enough.</i>
<i>Tier 2 search</i>	<i>Physics ablation and tuned direct/Tanh settings.</i>	<i>R2 = 0.7700;</i> <i>MAPE = 37.22%</i>	<i>Better than Tier 1, but still below the grid baseline.</i>
Tier 2b / current best	Residual target on leakage-free GridInterp base, SiLU activation, [16,8], 10-seed ensemble.	R2 = 0.8474; MAPE = 20.44%	Best current PINN-style result. Physics penalties added almost no extra benefit after the strong table base was included.

The main tuning lesson was that the model should not be selected using the real high-pressure test rows. The proxy search used lower pressures for training and the

15,000-16,000 kPa band for validation; only the final selected configuration was evaluated at 17,000-21,000 kPa.

10. External testing data from other papers

To test whether the model behaves sensibly in conditions that are different from the lookup table, data was collected from additional CHF papers. The project plan uses a precision order: appendix tables first, vector-figure extraction second, classical image processing third, and visual-language tools only for reading labels and legends.

Source and extraction	What is available
<i>Appendix tables</i>	<i>156 exact water helical-coil rows and 55 exact R123 straight-tube rows were identified, giving 211 rows that can be transcribed without guessing coordinates.</i>
<i>Values read from graphs</i>	<i>The inclined-tube paper gives approximately 10 fully specified points plus additional critical-quality points; these must be checked against axes and figure captions.</i>
<i>Parity plots</i>	<i>The R123 helical-coil paper has about 140 CHF points in parity plots, but many input features cannot be recovered reliably, so these are kept separate or marked as inferred.</i>

The external rows contain more information than the three-feature LUT, such as:

Feature group	Examples
<i>Core</i>	<i>Pressure, mass flux, quality, CHF.</i>

<i>Geometry</i>	<i>Tube diameter, heated length, L/D, coil diameter, pitch, inclination, straight/helical geometry.</i>
<i>Fluid and thermal state</i>	<i>Fluid name, reduced pressure, density ratio, surface tension, latent heat, inlet subcooling, Jakob or boiling number where available.</i>
<i>Provenance</i>	<i>Source paper, table or figure number, extraction method, measured/inferred flag, and quality-control notes.</i>

These external papers are being used as test-only data, not as training data. Water and R123 will be reported separately because changing the fluid and geometry creates a real distribution shift. This helps us see whether the PINN degrades more smoothly than tree models when the conditions change.

11. Work currently underway

- *Writing of the research paper has started. The literature review, validation protocol, model comparison, and PINN results are being organized into the paper.*
- *A CAD model design is also underway. The aim is to build a small experimental setup so that we can study CHF under different conditions ourselves and understand the physical concept alongside the ML model.*
- *Next steps are to finish quality checks on the paper-derived data, complete the external test comparison, and connect the CAD design with the conditions that can be measured experimentally.*

12. Main points for the professor

1. *The lookup table was cleaned carefully and reduced to **11,088** usable rows.*
2. *Split A and Split B show interpolation performance; Split C is the important high-pressure extrapolation test.*
3. *Tree models are excellent inside the known range but fail to extrapolate.*
4. *The current best trained approach is a residual-on-lookup-table SiLU ensemble with Split C **R2 = 0.8474**.*

5. External paper data and a CAD experiment are being added so the final work can be tested beyond the original three-feature water table.

All numerical values in this update come from the project repository result tables, PINN outputs, and the senior-review PLAN.md.

Project charts

Figure 1. R2 comparison across validation splits

